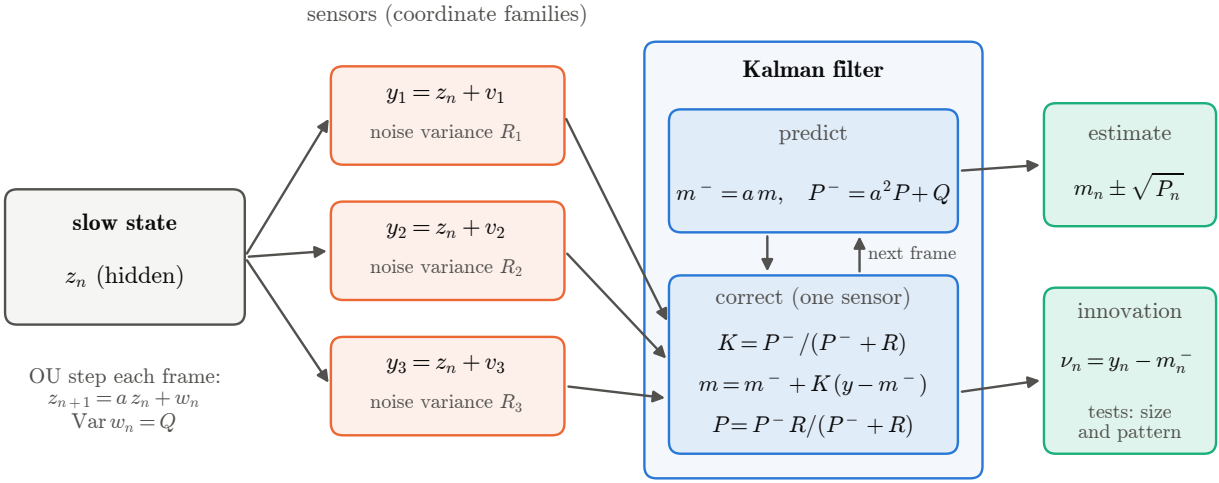
sensors (coordinate families)
slow state
z_n (hidden)
OU step each frame:
$z_{n+1} = a\, z_n + w_n$
$\mathrm{Var}\, w_n = Q$
$y_1 = z_n + v_1$
noise variance R_1
$y_2 = z_n + v_2$
noise variance R_2
$y_3 = z_n + v_3$
noise variance R_3
Kalman filter
predict
$m^- = a\, m, \quad P^- = a^2 P + Q$
next frame
correct (one sensor)
$K = P^-/(P^- + R)$
$m = m^- + K(y - m^-)$
$P = P^- R/(P^- + R)$
estimate
$m_n \pm \sqrt{P_n}$
innovation
$\nu_n = y_n - m_n^-$
tests: size
and pattern